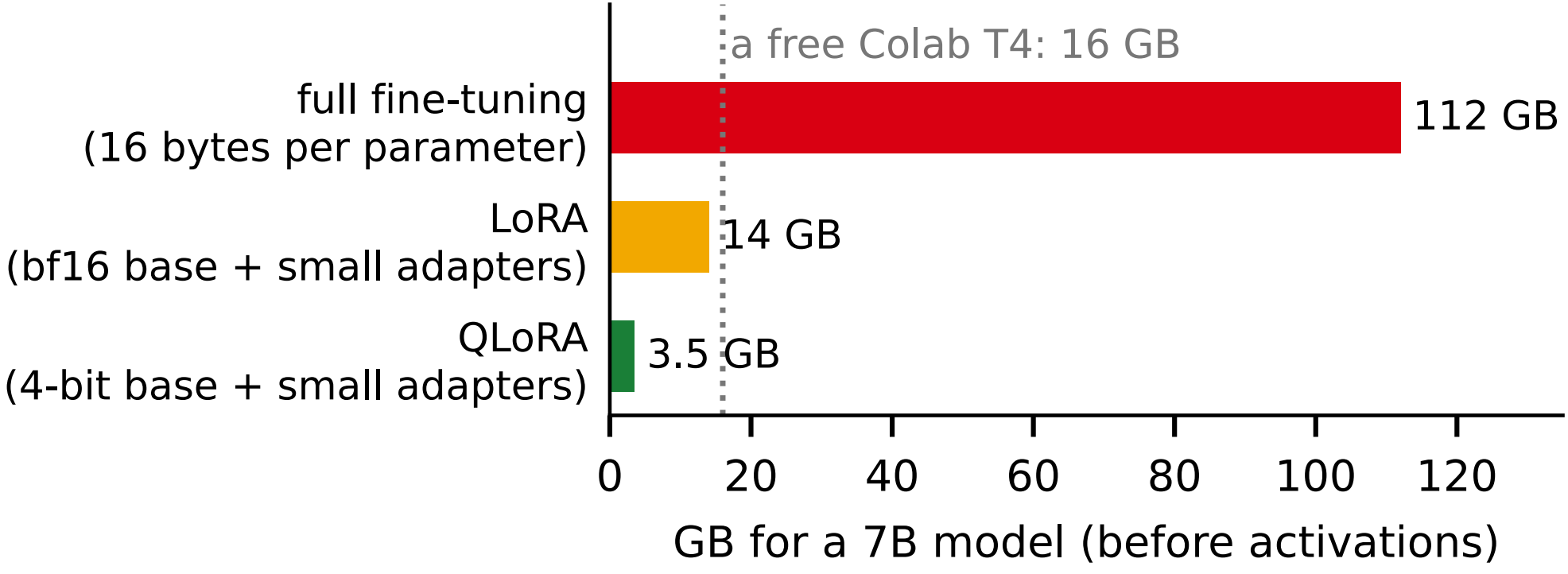
a free Colab T4: 16 GB
full fine-tuning
(16 bytes per parameter)
112 GB
LoRA
(bf16 base + small adapters)
14 GB
QLoRA
(4-bit base + small adapters)
3.5 GB
0
20
40
60
80
100
120
GB for a 7B model (before activations)